

Technical Proposal for Vehicle-to-Grid Integration of Berlin's Electric Bus Fleet



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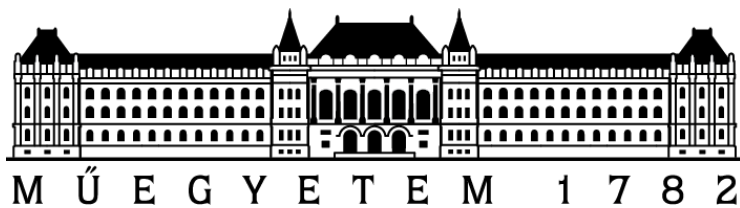


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1. Executive Summary

This project evaluates the techno-economic feasibility of implementing Bus-to-Grid (B2G) functionality at a BVG electric bus depot in Berlin. Since the fleet's electrification is already mandated and underway, the key question is whether upgrading planned charging infrastructure to bidirectional capability can provide additional operational and societal value. Rather than treating B2G as a standalone profit-making investment, the analysis frames it as a strategic enhancement to an unavoidable electrification process.

Our assessment confirms that energy arbitrage revenues alone cannot recover the full capital and operational costs of bidirectional chargers, an expected outcome for public transport operators that are not profit-oriented. However, when viewed through a municipal and system-value lens, B2G offers substantial benefits. The incremental cost between unidirectional and bidirectional charging - approximately 30 % per charger - is modest relative to the overall depot electrification investment. In return, B2G enables smart charging, peak shaving, and controlled load management, potentially reducing OPEX, lowering contracted grid capacity, and avoiding or deferring transformer and grid reinforcements.

Beyond cost savings, B2G supports Berlin's broader energy transition. By shifting charging into low-price, renewable-rich hours and discharging during system peaks, electric buses act as distributed flexibility assets that help balance the grid and reduce reliance on fossil peak plants. Unlike ancillary services, which are operationally incompatible with bus schedules, energy arbitrage fits naturally with depot return patterns and daily charging operations.

In conclusion, while B2G is not financially profitable when assessed solely through arbitrage revenue, it delivers significant operational, environmental, and system-level benefits. Given that Berlin must electrify its fleet regardless, integrating bidirectional charging into future depot upgrades represents a strategic enhancement to the city's electrification pathway.

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2. Introduction & Problem Statement

Cities across Europe are rapidly electrifying public transport fleets as part of broader climate and air-quality commitments. Berlin's public transport operator, BVG, is undergoing one of the largest e-bus rollouts in Germany, requiring substantial investment in depot charging infrastructure and grid connection capacity. This transformation introduces new challenges for the local electricity system, including higher peak loads, increased dependence on renewable generation patterns, and the need for smart integration of charging processes.

In this context, Bus-to-Grid (B2G) concepts are gaining attention as a way to enhance the value of the electrified fleet. By enabling controlled, bidirectional power flows between buses and the grid, depots can shift charging into low-cost, renewable-rich periods and discharge energy during high-demand evening peaks. Although B2G is not primarily a profit-driven technology, it offers significant potential to reduce operational expenditures, support local grid stability, and improve the overall efficiency of the electrification effort.

This project develops a techno-economic assessment of B2G for a representative BVG depot. It evaluates realistic bus availability, usable battery capacity, and German electricity market price patterns to quantify how much flexibility a depot can provide and what financial or operational benefits may result. Because the electrification of the fleet is unavoidable and already financed, this study focuses on the incremental value of upgrading from unidirectional to bidirectional charging — rather than on the profitability of arbitrage alone. The goal is to provide a pragmatic, system-oriented proposal that supports Berlin's smart-grid ambitions and enhances the long-term sustainability of the public transport system.

3. Literature Review

This section gives a short overview over the basics to understand the functionality and significance of the B2G business proposal.

3.1 Smart Grid & V2G

Smart grids represent the next evolution of electricity systems, integrating digital communication, distributed generation, and real-time control to manage increasing shares of variable renewables. Fang et al. (2012) emphasize that smart grids enable bi-directional

power flows, active demand-side participation, and improved system reliability. Gharavi and Ghafurian (2011) further highlight that smart grids are designed to accommodate decentralised energy resources and the operational uncertainty of wind and solar generation.

Within this framework, Vehicle-to-Grid (V2G) technology allows electric vehicles to act as distributed storage units that can discharge energy back into the grid. Kempton and Tomic (2005) established the foundational V2G concept, demonstrating its potential for frequency control, peak shaving, and energy shifting. Guille and Gross (2009) developed the conceptual architecture for V2G integration, outlining the necessary communication and market structures. Recent reviews show that while V2G faces economic and regulatory challenges, it can provide valuable system flexibility when intelligently scheduled (Sovacool et al., 2018).

Fleet applications, especially buses, are often cited as promising early adopters due to centralized charging, predictable schedules, and large aggregated battery capacities. Studies show that bus depots offer meaningful load-shifting potential, supporting both grid stability and operational cost reduction. More broadly, distributed flexibility from EVs is increasingly recognized as a key tool for integrating higher shares of renewable energy. (Jahic, 2022)

Together, the literature positions V2G as a technically feasible and strategically valuable component of smart grids, particularly in urban contexts with growing electric bus fleets.

3.2 Berlin's Bus Fleet Electrification Trends

Berlin's shift toward an electric bus fleet has accelerated in recent years. BVG currently operates 227 battery-electric buses, making it one of Germany's largest e-bus fleets (BVG, 2025). To expand this transition, BVG ordered 50 Solaris Urbino 18 electric buses in 2023 and subsequently placed a much larger order for 270 additional articulated e-buses, part of a framework contract allowing up to 700 vehicles over the coming years (Solaris, 2024; Randall, 2024).

Strategically, BVG aims to electrify around one-third of its ~1,500 buses by 2027, corresponding to roughly 500–550 e-buses on the road. Infrastructure development is progressing in parallel: construction of new e-bus depots and charging-capable workshops began in 2025, signalling long-term institutional commitment. (Prößer, 2025)

Operational data confirms the scale-up: BVG reports that e-buses covered 8.16 million kilometers in 2024, a 20% increase year-on-year (BVG, 2025). Public statements indicate the ambition to operate a locally emission-free bus fleet by 2030 (Sustainable Bus, 2024).

Overall, current literature and public data reveal a clear trajectory: rapid fleet expansion, major procurement commitments, significant infrastructure investments, and a strong political mandate for full electrification.

3.4 Germany's electricity demand

Germany's electricity market is increasingly shaped by high renewable penetration, which drives strong short-term price volatility. In 2024, renewables supplied over 60% of public net electricity generation, causing frequent low-price periods during night-time or solar peaks and high prices during evening demand peaks. Day-ahead prices commonly range from 20–50 €/MWh in low-demand hours to 150–200 €/MWh in peak hours, with occasional spikes near 300 €/MWh (SMARD, 2025). This volatility forms the economic basis for energy arbitrage, enabling flexible assets like e-bus depots to reduce charging costs and shift load. (Fraunhofer ISE, 2025; SMARD, 2025)

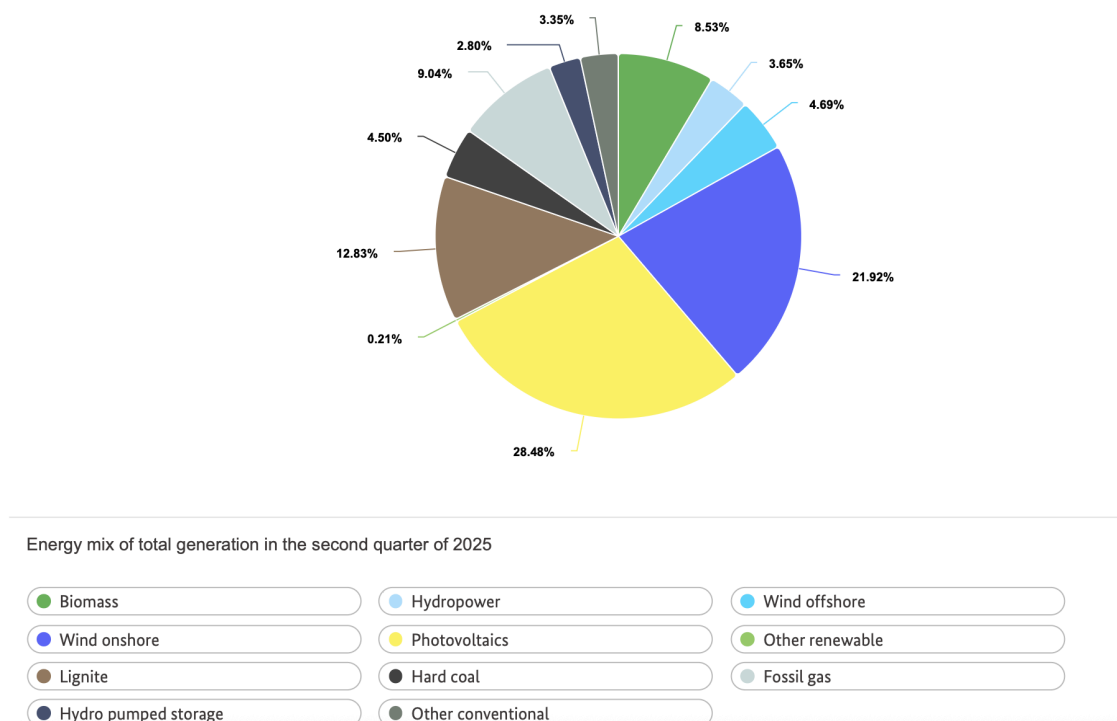


Figure 1. Germany' electricity production in the second quarter of 2025

4. Methodology

The current project aims to evaluate the potential contribution of BVG's electric bus fleet to Berlin's energy system through Vehicle-to-Grid (V2G) integration. The process to define the technical feasibility of the study is the following:

4.1 Bus Capacities of BVG

The first step aims to identify the current status of the e-bus fleet: how many units are currently in operation and which models are being used. This information will help determine the available energy storage potential of the fleet, which, within a certain range, could be integrated into the grid.

The baseline scenario will assume that all e-buses in the fleet are either Ebusco 2.2 or Solaris Urbino 18, since these are the most recently ordered models by BVG (BVG, 2025; Prößer, 2025)

Table 1: Main Characteristics of Ebusco 2.2 and Solaris Urbino 18 (Ebusco, 2025) and (Solaris Bus & Coach, 2023).

E-Bus Model	Battery Capacity	Length
Ebusco 2.2	> 400 kWh pack capacity > 500 kWh pack capacity long-range	12 m (single-deck)
Solaris Urbino 18	800 kWh	18 m

4.2 Bus Schedule Analysis

The next step involves examining the daily operational schedules of the e-buses, including their departure and arrival times at depots, using studies from other cities as reference. This analysis will support the estimation of how many units could realistically be available to participate in the V2G process.

The study quantifies the flexibility potential of electric-bus depots under unidirectional (depot-only) charging, using real schedules from two Hamburg depots (BD1 $\approx$ 26 MW max, BD2 $\approx$ 14 MW). Buses complete mainly one trip per day (leaving between 05:00 to 08:00 and returning 16:00 to 24:00 so depot activity is highly time-dependent. (Jahic et al., 2022)

Two charging modes were modelled:

1. Uncontrolled: buses charge immediately upon return, which is only negative flexibility (load reduction).
2. Controlled (peak-shaving): charging delayed or shifted, which is positive and negative flexibility.

Key Takeaways:

- Highest bus flexibility/availability: 16:00 to 24:00 (most buses parked; loads shiftable for less than 4 h).
- Lowest flexibility: 02:00 to 08:00 (pre-conditioning + departure times) and 08:00 to 16:00 (buses on road).
- Weekend days: Fewer trips lead to higher depot availability (1.5× higher flexible share can be assumed).
- Controlled charging lowers load peaks (27%) and adds +2 MW extra flexible power (BD1).
- Increased ambient temperature or no pre-conditioning lead to larger flexibility windows.

In conclusion, even with one-directional charging, depot-connected e-buses can provide several megawatts of shiftable load, representing approximately between 35 to 40% of the fleet available during evening hours. This quantification approach serves as a valuable basis for assessing grid service capabilities and the potential for vehicle-to-grid (V2G) integration (Jahic et al., 2022; Perumal et al., 2022).

4.3 Electricity Demand Profile in Berlin

To analyze Berlin's energy demand especially in peak demand times energy demand graphs from (Straub et al., 2023) is used and approximated. The paper analyzes the energy demand for 1 sub-district in Berlin. This paper approximates the energy demand in the peak evening hours for winter and summer, scales the energy demand up to the whole city and assesses the effectiveness of B2G off of that. Figure 2 shows the demand curves for a sub-district in Berlin during winter and summer.

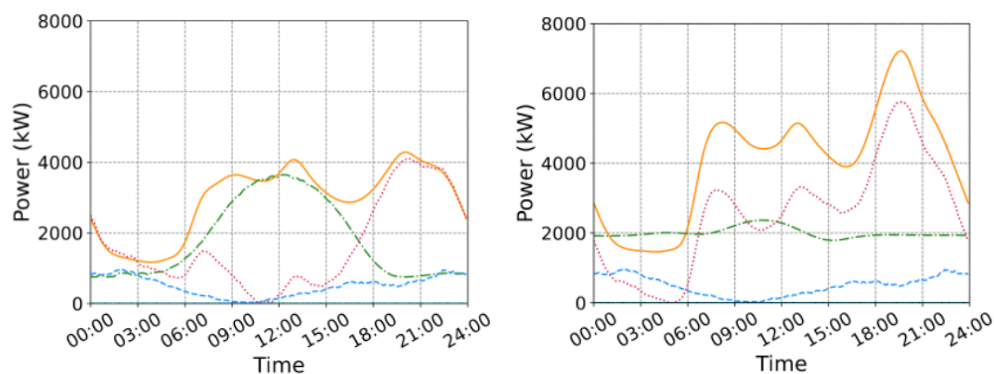


Figure 2. Electricity Demand in Berlin in Summer (left) and Winter (right)

4.4 Scenario Definition

Finally, future scenarios will be developed under the assumption of a relatively constant electricity demand curve for both winter and summer, while taking into account the planned increase in the size of the e-bus fleet.

The analysis will consider a unidirectional charger case and a bidirectional charger case. Bidirectional chargers have a higher CAPEX than unidirectional chargers, thus, the analysis will try to define if an extra investment on bidirectional chargers is justified from a revenue + savings point of view.

5. Impact Assessment

The proposed project offers benefits to the city of Berlin and its constituents. Figure 3 summarizes the different positive impacts of the project.

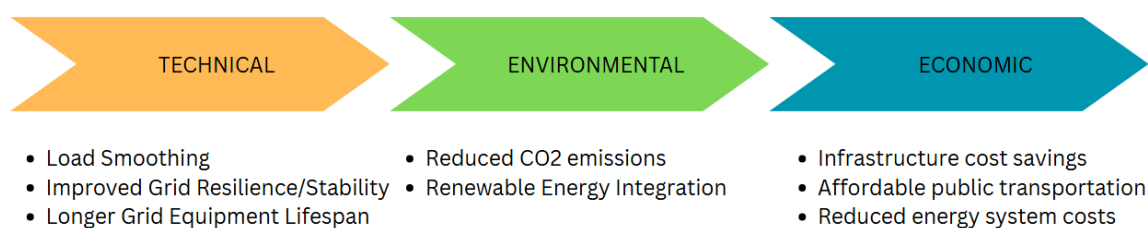


Figure 3. Technical, environmental, and economic impacts

5.1 Technical Impact

The addition of V2G buses with bidirectional charging features benefits the overall reliability of the grid through load smoothing, providing backup energy and reserves to make the grid stable, and prolonging the equipment lifetime of ancillary equipment such as transformers (Zhang et al., 2025).

With V2G, parked EVs or buses can absorb excess power during low-demand periods and supply power back during peak demand, evening out the load curve and reducing peak loads. This helps avoid sudden demand spikes and smooths overall electricity consumption throughout the day. The considered fraction of V2G buses parked during peak demand time, 15%, can discharge nearly 30MWh per day while most of the fleet charges during low demand times during the night.

The bus fleet in this project can function also as distributed energy storage in times of grid stress. In power outages they can feed electricity back, supporting critical loads, stabilizing frequency and voltage, and offering redundancy that enhances overall resilience. Consequently, this relieves electrical equipment load and protects them with unnecessary

surges that can affect the overall lifetime of the equipment. Prolonging the operational use of devices throughout the grid is expected to reduce the operating and maintenance expenses of the grid as well as lessen the downtime caused by maintenance and replacement of these electrical devices.

5.2 Environmental Impact

The fleet of 1,250 of V2G buses by 2030 can avoid up to 101 tons of carbon dioxide emissions per day. The reduced CO₂ emissions are created by displacing fossil-fuel “peak” generation (Zhao et al., 2022). This means when implementing V2G buses in Berlin, stored clean or off-peak electricity in EVs can be used during peak demand instead of firing up fossil-fuel peaker-plants, cutting carbon output. The prevented emissions of carbon dioxide are highly reliant on the energy mix of the electricity source for charging the V2G buses and further investigations using data Berlin’s energy mix would be necessary to further refine this estimated environmental impact. Moreover, the adoption of V2G bus technology also allows better integration of renewables as the buses can act as energy storage when there is a surplus from intermittent renewable energy sources such as solar and wind. This stored energy can then be discharged to the grid when needed.

5.3 Economic Impact

V2G technology for Berlin’s bus fleet acts as a dual purpose investment that would not only serve as a cleaner and cheaper mode of transportation in the city but also as battery energy storage service for the local grid. This drastically lowers the infrastructure costs for Berlin’s transportation and energy storage.

For grid infrastructure costs, recent study estimates that grid operators across Europe could save about €4 billion per year thanks to smart charging that would necessarily lead to lowering of network stress and delaying or avoiding infrastructure upgrades. This type of savings could be translated into the city-scale context for Berlin leading to more economic benefits from this V2G bus fleet technology. An electric bus fleet running on V2G would greatly reduce the operating expenses for both the city government and the public-transport operators.

In the case of Berlin, the project also aligns well with the city’s energy objective. Especially in efficient energy use. This also translates to satisfying Germany and Europe’s goals of reducing energy wastes. A recent report predicts that bidirectional charging could reduce

annual EU energy-system costs by up to €22.2 billion by 2040. Through the proposed project, Berlin takes a step closer to a cleaner and more energy-efficient economy.

6. Financial Plan

As previously mentioned, the project's scope considers the scalable acquisition of E-buses for BVG, the development of charging stations as well as additional infrastructure to cope with the requirements of the fleet, the investment on chargers and the recognition of key charging and discharging windows in Berlin. The nature of the project is to estimate the requirement investment needed to cope with the public demands of a public transport system which requires it to be fully aligned with Germany's energy transition and to optimize operative expenditures (OPEX).

Taking this into account, the financial plan was split into two main cases: Investment in unidirectional charging, which will allow the optimization of charging windows, and investment in bidirectional charging, which will allow the optimization of charging windows as well as the sale of electricity during specified discharging windows.

6.1 CAPEX

Key assumptions for the CAPEX development where:

1. The scalable acquisition of e-buses to comply with the objectives established by BVG (227 by 2025, 550 by 2027 and 1.250 by 2030).
2. An estimated bus price of 550,000 per unit considering that the acquired models will be Ebusco 2.2 and Solaris Urbino 18.
3. The development of Charging Stations considers the acquisition of chargers, depot upgrades, software installation, implementation of communication systems and civil costs.
4. Taking into consideration the projected cost of smart chargers and bidirectional prices from 2025 to 2030 (Vollmoth et al., 2024), a heuristic was developed to link the required investment per KW provided. This heuristic is €350/kW for unidirectional chargers and €455/kW for bidirectional chargers. Bidirectional chargers have a 30% higher CAPEX than unidirectional chargers.

The estimated CAPEX for the 2025-2030 period was between 699 and 740 Million Euros considering either Unidirectional or Bidirectional cases. The only difference in these cases was the investment cost considering the development of Charging Stations.

Unidirectional

Unidirectional chargers offer a simpler and more robust configuration, reducing hardware requirements, control strategies, and communication interfaces compared to bidirectional (V2G or V2H) systems. Since the case focuses on supplying energy to the battery and optimizing charging windows rather than returning it to the grid, adopting a unidirectional topology avoids additional investment, regulatory considerations, and interoperability challenges.

***Table 2.** Unidirectional CAPEX for Period 2025-2030*

Consideration	Period 2025-2030
E-Bus Fleet Acquisition	€562,650,000
Charging Stations Investment	€136,500,000
Total CAPEX	€699,150,000

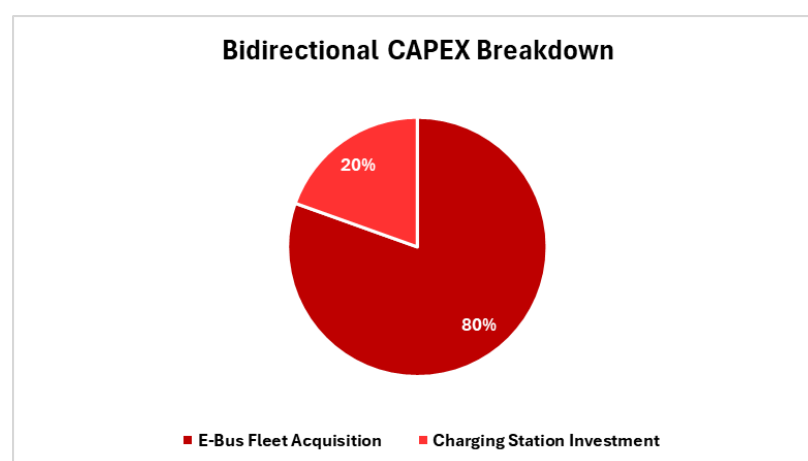


Figure 4. Unidirectional CAPEX Breakdown

Bidirectional

Bidirectional chargers were selected for this scenario because they enable both charging and controlled energy injection back to the grid, unlocking additional flexibility and system value that unidirectional solutions cannot provide. Although bidirectional systems require more advanced hardware, communication capabilities, and control algorithms, the added functionality supports vehicle-to-grid (V2G) or vehicle-to-home (V2H) strategies, improves peak-shaving potential, and enhances overall grid resilience. The effect of the higher investment costs can be seen in an overall increase of the CAPEX between 5 and 10%.

Table 3. Bidirectional CAPEX for Period 2025-2030

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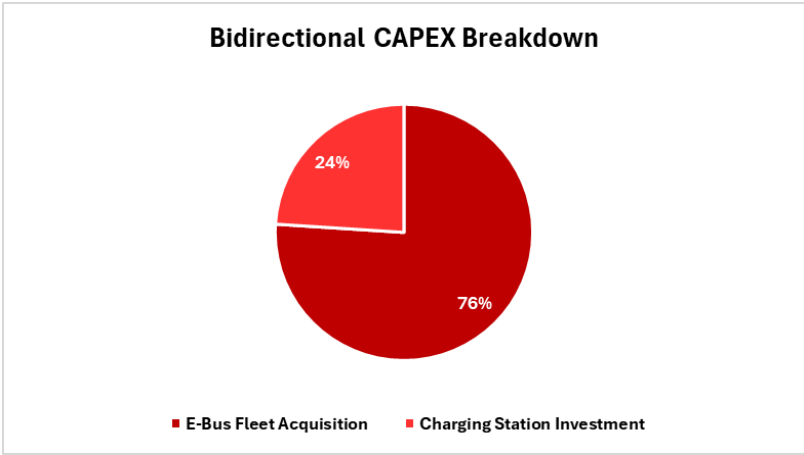


Figure 5. Bidirectional CAPEX Breakdown

6.2 OPEX

The developed OPEX is considered for both unidirectional and bidirectional cases, since they have a baseline charging price that eventually will be compared with lower prices.

Key assumptions for the OPEX development where:

1. O&M from E-Buses and Charging Stations is 15% of the total CAPEX required, respectively.
2. Availability of 50% of the E-Bus fleet to be charged between 00:00 and 05:00.
3. The average battery capacity considered is 520 KWh, since it's a middle ground between the battery capacity of Solar Urbino 18 and Ebusco 2.2 (Solaris Bus & Coach, 2023) .
4. The charge load is 80% of the battery's capacity.
5. The baseline charging price is €90/MWh (SMARD, 2025).

Table 4. Baseline OPEX for Period 2025-2030

Consideration	Period 2025-2030
E-Bus O&M	€276,705,000
Charging Stations O&M	€90,499,500
Total Charging Expenses	€114,586,056
Total CAPEX	€481,790,556

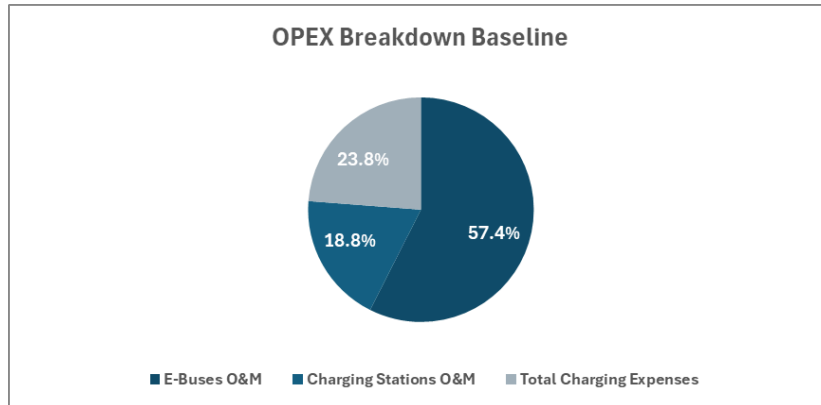


Figure 6. OPEX Breakdown

The estimated OPEX under baseline conditions is around 481 Million Euros, with a high reduction potential in Charging Expenses considering lower charging prices due to the optimized window (00:00-05:00).

6.3 Revenue Model

The main revenue stream considered in the project, exclusively for the bidirectional case, was the active sale of electricity by discharging batteries of unused buses during peak times. Peripheral revenue streams such as ticket sales or advertisement aren't considered within the project's scope.

Key assumptions for the revenue model development where:

1. Availability of 15% (Scenario A) of the E-Bus Fleet to be discharged between 06:00-09:00 and 16:00-21:00.
2. The considered discharge load is 50% of the battery's capacity.
3. The baseline discharging price is €150/MWh after considering a conservative approach from the maximum price of €280/MWh (SMARD, 2025).

With a scaling revenue proportional to the development of the E-Bus fleet, the total revenue that could potentially be produced battery discharging for the period 2025-2030 are 29.2 Million Euros. By the last implementation stage, the average revenue per bus would be around €10,250/Bus.

Table 5. Discharging Revenue per Scalable Period

Consideration	2025 to 2026	2027 to 2029	2030
Daily Discharging Capacity [MWh]	2.7	10.7	29.3
Daily Discharging Revenue	€3,187	€12,870	€35,100
Total Discharging Revenue	€2,326,568	€14,092,650	€12,811,500

6.4 Unidirectional Case Results with OPEX Sensitivity Analysis

A Sensitivity Analysis was done to simulate low charging prices which are coherent to the optimized charging windows. This analysis will substitute the baseline charging price of €90/MWh.

Table 6. Scenarios for OPEX Sensitivity Analysis

Type of Sensitivity Analysis	Considered Charging Price [€/MWh]
Scenario 1	10
Scenario 2	20
Scenario 3	30

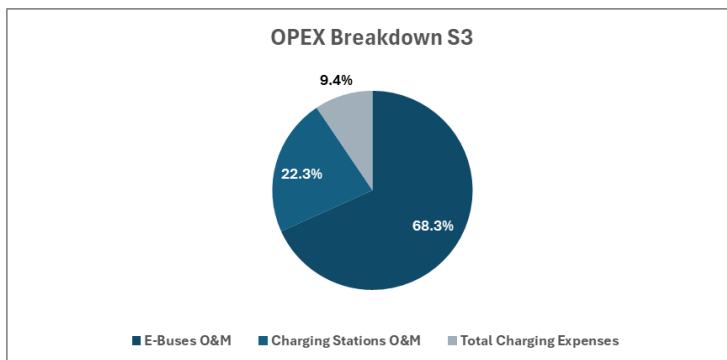
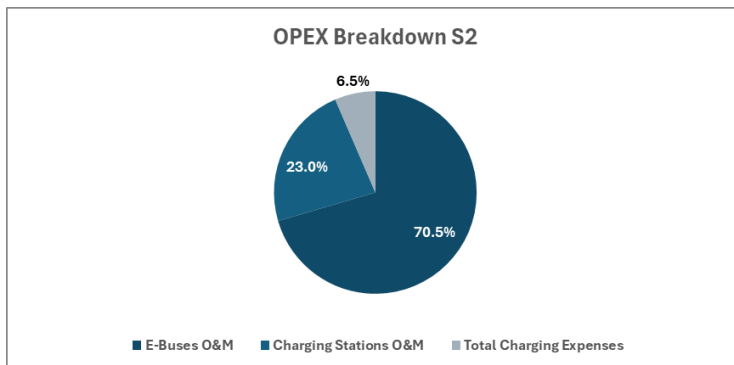
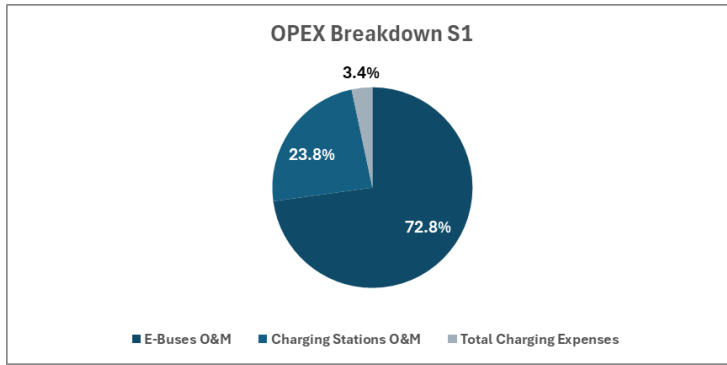


Figure 7. OPEX Breakdown of different scenarios

Total Charging Expenses range between 12.7 to 38.1 Million Euros compared to 114.5 Million Euros. Needless to say, Total Charging Expenses decrease when the considered charging price is lower.

When accounting the established CAPEX for the unidirectional case and OPEX Sensitivity Analysis a Total Savings per Sensitivity scenario was developed. This implies the difference between Total Charging Expenses of every sensitivity scenario against the baseline.

Table 7. Projected Savings per Sensitivity Analysis Scenario in Period 2025-2030

Type of Sensitivity Analysis	Projected Savings in Period 2025-2030
Scenario 1	€101,854,272
Scenario 2	€89,122,488
Scenario 3	€76,390,704

Figure 8. compares the magnitude of the potential Savings against the CAPEX and OPEX invested in each Sensitivity scenario.

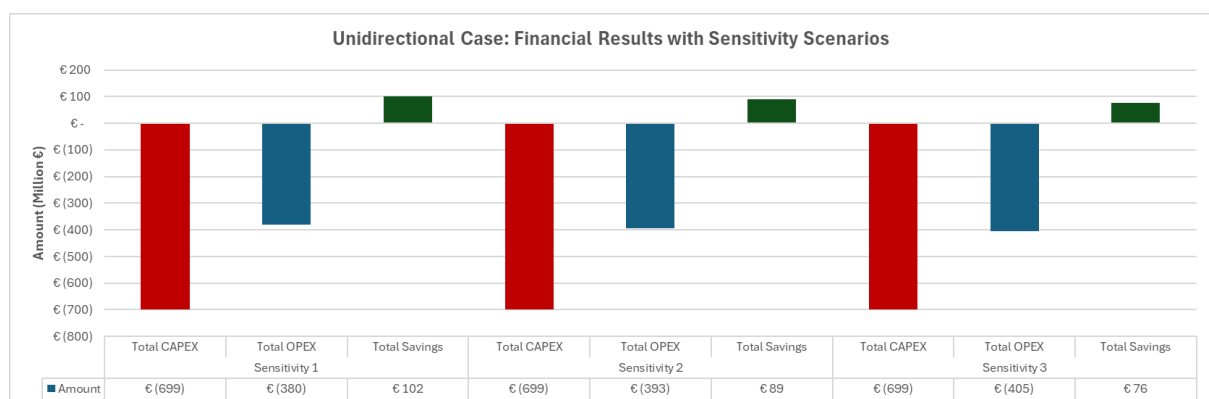


Figure 8. Financial Results by Sensitivity Scenario

To summarize the benefits of the project, the indicator Benefit in OPEX % was established. Which is a relationship of the sum of Savings and Revenue (applicable in the bidirectional case) divided by the Total OPEX of the baseline scenario. Figure 9 visualizes the benefit comparison by Sensitivity Scenario, achieving up to a 21% in the OPEX Scenario 1.

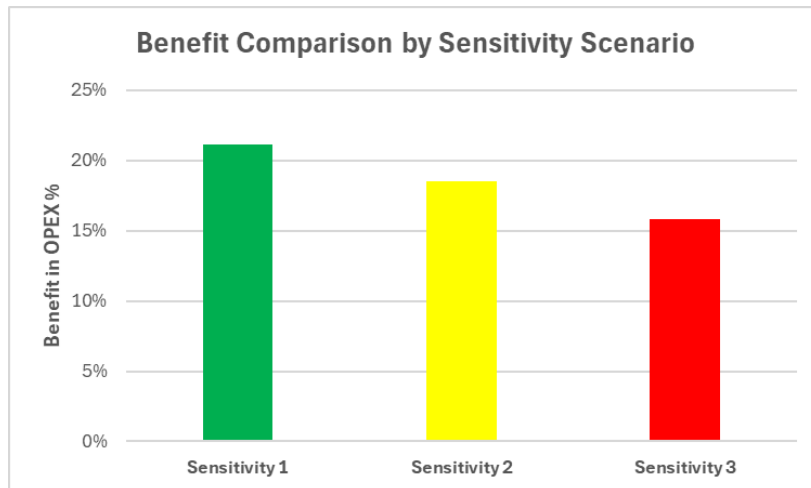


Figure 9. Benefit in OPEX% per Sensitivity Scenario

6.5 Bidirectional Case Results with OPEX and Revenue Sensitivity Analysis

The Sensitivity Analysis done for the Unidirectional Case was repeated in the Bidirectional Case, nevertheless, a second sensitivity analysis was included to simulate higher bus availability to discharge their batteries during high electricity demand hours.

Table 8. Scenarios for Revenue Sensitivity Analysis

Type of Discharging Scenario	Fleet Availability to Discharge in Peak Hours [%]
Scenario A	15%
Scenario B	25%
Scenario C	30%

Project revenue ranged between 29.2 and 58.5 Million Euros, depending on the chosen Discharging Scenario. Scenario C which has the highest fleet availability, thus a higher discharging capacity to support the grid, will bring a bigger revenue to BVG.

By combining all the possibilities of charging prices and discharging scenarios, a matrix was generated to understand which combination provided the biggest amount of Benefit in OPEX %. The bidirectional scenario 1C, which is the one with the highest discharging availability (30%) and lowest charging price (€10/MWh), provided a Benefit in OPEX % of 33%. For the

opposite scenario 3A, low discharging availability (15%) and highest charging price (€30/MWh), the Benefit in OPEX % is 22%.

Considers Benefit in OPEX % for the period 2025-2030

	Discharging A	Discharging B	Discharging C
Sensitivity 1	27%	31%	33%
Sensitivity 2	25%	29%	31%
Sensitivity 3	22%	26%	28%

Figure 10. Benefit in OPEX % for possible OPEX Sensitivity and Discharging Scenarios.

The conclusion of this exercise is that, by investing an extra 40 Million Euros to convert the chargers from the unidirectional case to the bidirectional case, Benefits in OPEX % can range between 22% to 33%, a considerable increase from the 16% to 21% range in the unidirectional case. The investment is easily justifiable considering the revenue range of 29 to 58.5 Million Euros per year.

6.6 Business Model Canvas

Figure 11 shows the business model canvas for a V2G-enabled electric bus system in Berlin. It highlights how the project creates value for the city, the grid operator, and the public transport authority. At its core, the value proposition focuses on reducing operational costs for bus depots, strengthening grid stability during peak hours, and supporting Berlin's climate-neutrality goals by enabling clean, flexible energy services. The key partners include the local DSO (Stromnetz Berlin), bus manufacturers, charging-infrastructure providers, software and energy-management companies, and the BVG transit authority. These partners enable the key activities such as V2G operation, fleet charging management, maintenance, data integration, and participation in energy markets. The model relies on key resources such as the electric bus fleet, bidirectional chargers, depot infrastructure, and digital energy platforms. Customer segments include the municipality, the grid operator, and indirectly, Berlin residents who benefit from a more resilient and cleaner energy system. Revenue streams come from avoided grid fees, flexibility payments, peak-shaving services, energy arbitrage, and operational savings, while the main costs involve depot upgrades, charger

installation, energy-management software, and maintenance. Overall, the canvas shows that V2G for buses can be financially viable and strategically aligned with Berlin's long-term sustainability and grid-modernization goals, provided that technology reliability and regulatory clarity continue to improve.

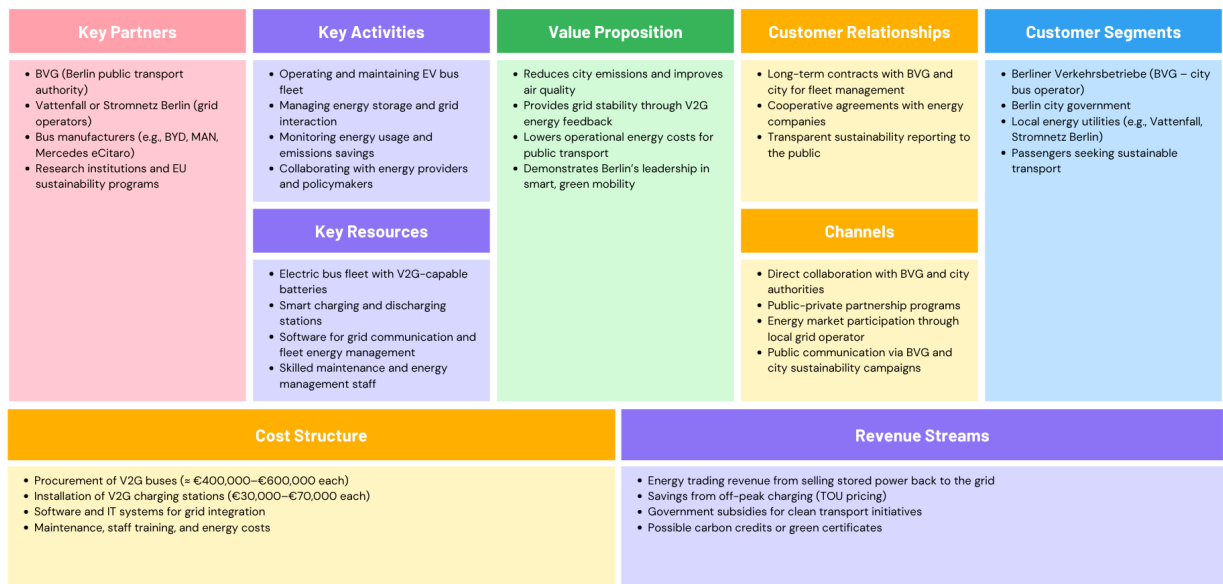


Figure 11. Business model canvas for the B2G project

7. Risk Analysis

The major potential sources of risks identified for this project and their corresponding mitigation strategies were formulated. Table 8 summarizes the risk analysis of this project.

Table 9. Risk analysis for V2G bus project

Risks	Mitigation Strategies
Technical Risks - Battery degradation - Reliability of chargers - Vulnerability of back-end systems	- Inclusion of battery life management strategies in planning such as proper power cycling and charging techniques - Installation of additional charging capacity and redundant chargers - Development of a robust and secure back-end system
Economic Risks	Long-term electricity price contracts for V2G

• Electricity market prices • Technology costs • Policy and regulation	bus fleet charging and discharging • Fixed price for the charger and installation costs from the suppliers • Align targets with known incoming legislations and engage policy formulation process as a key stakeholder
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7.1 Technical Risks

The main sources for technical risks for this project include faster battery degradation due to improper cycle management, complexity and technological maturity of bidirectional chargers that may lead to failures, and availability and vulnerability of backend systems to support V2G bus fleets operation. The variability of returns are highly dependent on the market conditions for both energy and technology (Eltohamy et al., 2025) To mitigate the potential effects of these risks, battery life management should be included in planning, redundant chargers or additional charging capacity in the bus depots should be available in case of failure, and ensure the development of robust backend services with sufficient protection from outside threats.

7.2 Economic Risks

For economic risks, the identified factors include electricity prices, technology prices, and legislation (Bai et al., 2022). The volatility of electricity-market prices driven by geopolitical and international influences which could lead to revenues lower than initially projected. The current cost of technology is also uncertain due to its fast development although price projections are expected to lower in the coming years. In terms of regulation, policies for V2G are still lagging. Preferred legislations for the adoption of this technology would require regulations in V2G metering and initial subsidization of V2G by the state. To reduce the possible impacts of these economic risks, strategies such as negotiating for long-term electricity price contracts and fixed price for charger purchase and installation will be necessary. Lastly, the planning and design phase of the project should align with future EV and V2G policies by working with the local and international regulators.

8. Final Recommendation & Roadmap

For future rollout, the next three phases are proposed: pilot, scale-up, and full-scale grid integration. The first step is to launch a pilot at a single depot with a limited number of

bidirectional chargers to validate technical performance, revenue potential, and operational impacts. This pilot should run for at least one full seasonal cycle to test winter and summer load conditions. Once validated, the program can scale up to additional depots, gradually increasing the share of V2G-enabled chargers while integrating automated charge-scheduling software. Finally, large-scale deployment should be incorporated into the city's long-term grid-planning strategy, allowing the DSO to use bus depots as predictable flexibility assets. This final phase includes formalizing B2G into grid-support contracts, locking in electricity and technology prices, preparation of grid such as upgrading of equipment for upcoming B2G integration, and integrating B2G's storage capacity to the city's electricity supply. Figure 12 summarizes the next steps from pilot to full scale deployment of Berlin's B2G system.

In addition to the features talked about in previous sections, the potential of integrating avoided-emissions data into BVG's public-facing communication for example, displaying "CO₂ saved through smart charging" on bus tickets or apps can strengthen public support for V2G by making the environmental benefits tangible to passengers. Opening selected V2G-enabled bus charging stations for limited public EV use could further enhance visibility while providing additional community value, positioning depots as neighbourhood energy hubs while providing a new revenue stream for the company. The combined impact of avoided grid-capacity charges, reduced peak-load penalties, and deferred transformer upgrades can generate operational savings that offset the incremental cost of bidirectional chargers. Together, these elements support a compelling narrative that V2G does not only strengthen Berlin's grid and reduces emissions but also offers a cost-effective enhancement to infrastructure the city must build anyway.

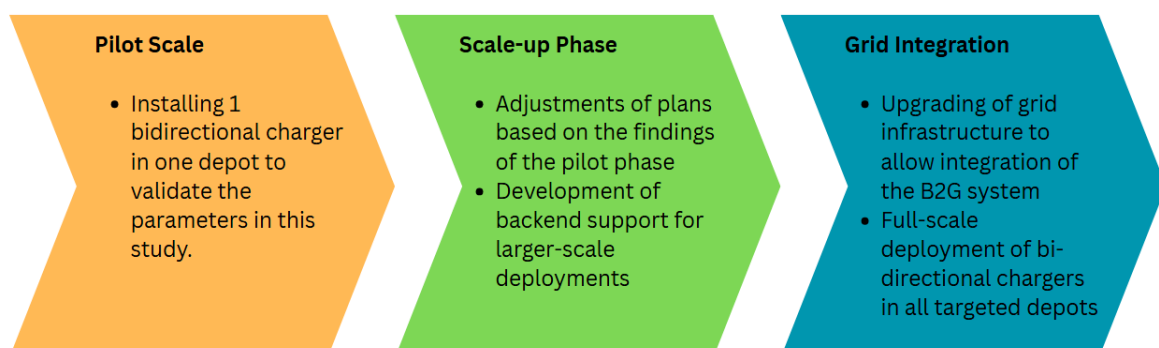


Figure 12. Implementation plan from pilot to full-scale B2G

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